

Mohamed Ali Lawini

Embedded Software Engineer

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Summary

Embedded Software Engineer with 5+ years of experience in low-level programming, firmware development, and hardware-software integration. Strong expertise in STM32, Embedded C/C++, Linux, and automotive embedded systems, with hands-on experience in Java native integration (JNI). Proven ability to design device drivers, debug complex systems, and improve software reliability.

Skills

- Programming: C, C++, Java, Embedded C
- Embedded: STM32, Firmware, Device Drivers, SPI, CAN
- Systems: Linux, AOSP, HAL, BSP, Kernel Development
- Tools: STM32CubeMX, Git, CI/CD, Docker, JUnit, gtest, gmock
- Concepts: Low-Level Programming, Embedded Systems, Debugging, Real-Time Systems, Hardware-Software Integration

Experience

Embedded & Tools Software Engineer

ACTIA ES

2023 – Present

- Delivered 10+ production features for STM32CubeMX ecosystem used by embedded developers worldwide
- Reduced defects by 30% via root-cause analysis and fixes
- Improved regression coverage by 40% using JUnit automation
- Contributed to architecture decisions and CI/CD improvements (20% faster builds)

Software Engineer (Android Automotive)

ACTIA ES

2021 – 2023

- Implemented FOTA updates for automotive infotainment systems
- Developed CAN kernel drivers and embedded C firmware
- Built C++17 services using modern design patterns
- Reduced debugging time by 25% using ADB, Traceview, Systrace
- Fixed SELinux issues ensuring system security and stability

Project

AI Short-Circuit Detection (EDF France)

ACTIA ES

Final Year Project — Jan – Jun 2021

- Built embedded AI system for high-voltage fault detection
- Designed SPI driver for STM32H745 with strict timing constraints
- Implemented ANN model on STM32 using STM32Cube.AI

Education

- Engineering Degree in Embedded Systems — FST Tunis (2021)
- Master's Degree in Electrical Engineering, Electronics and Automation — FST Tunis (2020)
- Bachelor's Degree (Licence) in Electrical Engineering, Electronics and Automation — FST Tunis (2018)

Certifications

ISTQB Certified Tester Foundation Level (CTFL) v4.0

Languages

Arabic (Native), French (Fluent), English (Professional)